Course Title: Mobile Application Development using React Native

Duration: 16 Weeks

Week 1 Introduction to Mobile App Development and React Native

Learning Objectives:

- Understand mobile app development paradigms (native, hybrid, cross-platform).
- Learn React Native basics and environment setup.

Key Skills:

- Node.js, Expo CLI installation
- Create basic React Native app

Practical Exercise:

- "Hello World" app using Expo

Week 2: React Native Components and Styling

Learning Objectives:

- Learn core React Native components and styling options

Key Skills:

- View, Text, Image, Button
- Inline styling and StyleSheet

Practical Exercise:

- Build a simple UI with header, image, and button

Week 3: State and Props

Learning Objectives:

- Understand state and props for dynamic behavior

Key Skills:

- useState hook, component communication via props

Practical Exercise:

- Counter app with increment/decrement
- Child component receiving props

Week 4: Event Handling and Forms

Learning Objectives:

- Handle user input, events, and form validation

Key Skills:

- onPress, onChangeText
- TextInput validation

Practical Exercise:

- Login form with username/password validation

Week 5: Navigation in React Native

Learning Objectives:

- Implement multi-screen navigation

Key Skills:

- React Navigation (StackNavigator, TabNavigator)
- Pass parameters between screens

Practical Exercise:

- Multi-screen app with at least 3 pages

Week 6: MIDTERM 1 – CRUD Operations with Local Storage

Learning Objectives:

- Implement basic CRUD functionality in React Native apps

Key Skills:

- Local data storage using AsyncStorage
- Create, Read, Update, Delete operations

Practical Exercise:

- Simple To-Do app with CRUD operations

Assessment:

- Midterm Practical Submission

Week 7: Working with Lists and Dynamic Data

Learning Objectives:

- Display and manage dynamic lists

Key Skills:

- FlatList, SectionList
- Key extraction and dynamic updates

Practical Exercise:

- Shopping list app with add/delete functionality

Week 8: APIs and Data Fetching

Learning Objectives:

- Fetch data from REST APIs

Key Skills:

- Axios, Fetch API
- Async/await and error handling

Practical Exercise:

- Build an app that fetches user data from a public API

Week 9: State Management with Context API / Redux

Learning Objectives:

- Global state management

Key Skills:

- Context API (Provider/Consumer)
- Redux store, actions, reducers

Practical Exercise:

- To-Do app with global state using Redux

Week 10: Advanced Components and Modals

Learning Objectives:

- Explore modals, alerts, and pickers

Key Skills:

- Modal, Alert, Picker components
- Nested navigation

Practical Exercise:

- Multi-screen app with modals and user interactions

Week 11: Device Features and Expo APIs

Learning Objectives:

- Access device hardware and features

Key Skills:

- Camera, Location, Notifications APIs
- Permissions handling

Practical Exercise:

- Camera app that stores photos locally

Week 12: MIDTERM 2 – Backend Integration

Learning Objectives:

- Connect React Native app with backend
- Perform CRUD operations with a database

Key Skills:

- Setup XAMPP with MySQL or .NET API backend
- REST API calls from React Native
- Fetch, post, update, delete data from backend

Practical Exercise:

- Build a CRUD app (e.g., Student Management) with API integration

Assessment:

- Midterm Practical Submission

Week 13: Advanced API Integration and Security

Learning Objectives:

- Secure API calls and manage authentication

Key Skills:

- Token storage in React Native

Practical Exercise:

- App with login, register, and protected routes using API

Week 14: Working with Third-Party Libraries

Learning Objectives:

- Use libraries for charts, maps, and analytics

Key Skills:

- React Native Maps, Chart libraries
- Integration of third-party SDKs

Practical Exercise:

- Different APIs like PHP , .Net

Week 15: Revision

Learning Objectives:

- Combine all learned concepts in a full-fledged app

Key Skills:

- Multi-screen app, state management, API integration, device features

Practical Exercise:

- Continue building the final project for evaluation

Week 16: Project Evaluation and Presentation

Learning Objectives:

- Present, demonstrate, and submit the final project

Key Skills:

- Full app development lifecycle
- Presentation, documentation, code quality

Assessment:

- Final Project Submission and Presentation
- Grading: Functionality (80%) and UI/UX (20%)